

Intro to Databases (5 days) :

Day 1 :

- DB concepts
- DB Design (ERD => Entity Relationship Diagram)
- File Based Systems
- DB Systems
- DB lifecycle

Day 2 :

- DB Mapping (Schema)
- Create DB
- Tool (SQL Server)
- SQL (Structured Query Language)

Day 3 :

- Joins
- Like
- Normalization

Day 4 :

- EERD (Enhanced ERD)
- Aggregate Functions
- Grouping
- Union
- Subqueries
- Execution Cycle

Day 5 :

- SQL Server Engine
- Transact-SQL (T-SQL)
- Top
- Select into
- Bulk insert
- Newid
- Security

Advanced SQL Server (5 days) :

Day 6 :

- DB integrity
- Constraints
- Rules
- SQL Server Schema and Security

Day 7 :

- Variables
- Conditions
- Loops
- Functions
- Parameters

Day 8 :

- View
- Index
- Pivoting
- Merge

Day 9 :

- Stored Procedures
- Triggers
- Jobs
- Script transaction

Day 10 :

- Cursor
- SQL CLR
- Snapshot

DB Life Cycle :

- 1- Analysis => (System Analyst or Business Consultant) Information gathering from the client. Produces the Requirement Document.
 - 2- DB Design => (DB Designer) Produces the ERD (Entity Relationship Diagram) , Note : Entities are NOT tables.
 - 3- DB Mapping => (DB Designer) Applies set of rules to produce the Database Schema , it's the actual tables that will be in the database
 - 4- DB implementation => (DB Developer) make the database it self , the physical database : physical files on the computer [.mdf => Contains data and metadata] and [.ldf => Contains logs and history and transactions that are done on the database]. We here use a RDMS (Relational Database Management System, ex: SQL Server [Microsoft] , Oracle , MySQL , Access ...) to create the DB to have a centralized place to put the data on (DB Sharing). So when we work in a company we will connect to the server of the centralized database , not like now each student has a server for him on his device
 - 5- Application => (Application Programmer OR DB User) Develops the application and the GUI (graphical user interface) that makes interacting with the database easier , through any technology (mobile , web , ...)
 - 6- Client => End User , uses the DB through the application developed (web app , mobile app , ...)
- Data VS Metadata :
- o Data : Actual data inside the database tables
 - o Metadata : column names , relationships names , table names, ...

Can we have an application without database ? YES : Static web site , standalone application , small mobile application

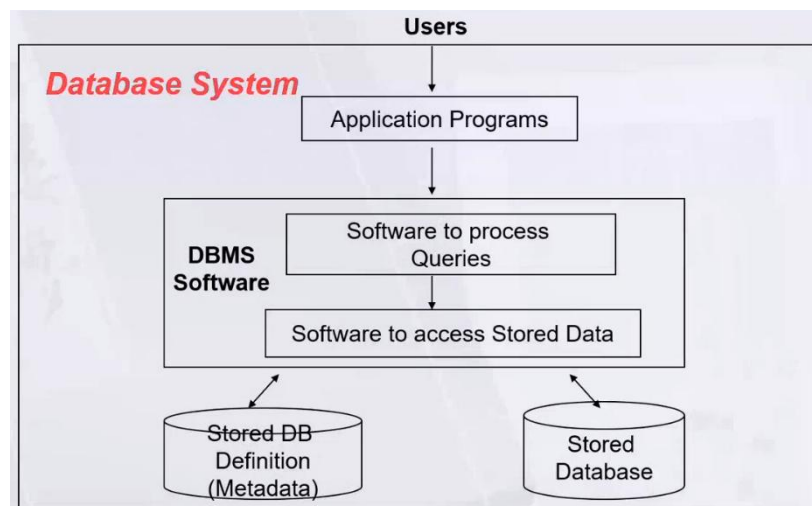
Before Databases , File Based Systems were used

- 1- Delimited File : data are separated by a delimiter (, / .)
- 2- Fixed-Width File : Fixed number of bytes for each attribute , ex: Id is 2 bytes , name is 5 bytes , ...

Delimited File	Fixed Width File																																
<table><tr><th>Id</th><th>Name</th><th>Age</th><th>DeptId</th></tr><tr><td>1</td><td>ali</td><td>22</td><td>10</td></tr><tr><td>2</td><td>Amr</td><td>32</td><td>20</td></tr><tr><td>3</td><td>Ahmed</td><td>30</td><td>30</td></tr><tr><td>1</td><td>ali</td><td>22</td><td>10</td></tr><tr><td>6</td><td>mai</td><td>25</td><td>100</td></tr></table>	Id	Name	Age	DeptId	1	ali	22	10	2	Amr	32	20	3	Ahmed	30	30	1	ali	22	10	6	mai	25	100	<table><tr><th>DeptId</th><th>DeptName</th></tr><tr><td>10</td><td>Sales</td></tr><tr><td>20</td><td>HR</td></tr><tr><td>30</td><td>Marketing</td></tr></table>	DeptId	DeptName	10	Sales	20	HR	30	Marketing
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File system Disadvantages and Limitations :

- 1- Duplication of data
 - 2- No sharing of data between users , each user with a separated copy of data (Reading inconsistency)
 - 3- Difficult Searching (linear => Full File Scan , Low Performance)
 - 4- No Constraints (ex: age must be more than 30) , so more app development time and complexity
 - 5- No standardization or data quality or integrity
 - 6- Program data dependance (no standard file format) , maybe the data from each file are in different formats so the program must maintain metadata for each file
 - 7- Manual backup and restore
 - 8- No security and permissions
 - 9- No relationship between files , so some invalid data can appear
-
- Database : A collection of related data , data + metadata , (in SQL Server => mdf and ldf files)
 - Database Management System : A software package / software system used to create and maintain computerized databases, then RDBMS (Relational Database Management System) appeared in 1980 .
 - Database System : The DBMS software together with the data itself. Sometimes the applications are also included (Software application + Database)



Database Systems advantages :

- 1- Standardization (SQL), better data accessibility and response
- 2- Relationship between tables
- 3- Sharing data between multiple users
- 4- Enforcing constraints and rules => integrity
- 5- Security and restricting unauthorized access
- 6- Providing backup and recovery

- 7- Minimum data redundancy because of using Primary Keys (Unique and not null value)=> increases data integrity consistency
- 8- Data Quality => column has a datatype and a defined size and data validations
- 9- Program data independence => metadata stored in DBMS , app doesn't worry about data formats

Database Disadvantages :

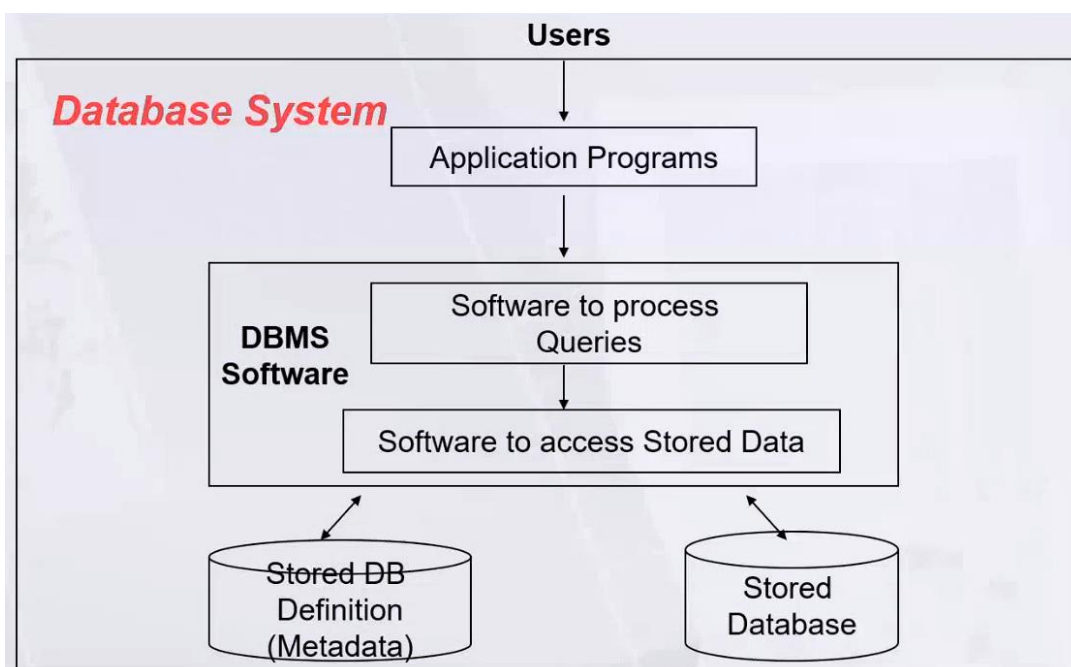
- High license cost
- Expert users
- Incompatibility between multiple DBMSs

Database Users :

- Database Administrator : Setup and configuration of the DB , Security and providing users and logins for database users , Backup and restore and recovery of the database.
- System analyst or Business Consultant : getting information from the client (ex: HR system analyst , Banking system analyst , ...)
- Database Designer : Does ERD and Schema
- Database Developer : Knows SQL and how to create , write queries and how to deal with different database objects.
- Application Programmer : Interface for using the DB , GUI , ..
- BI developer or Big data specialist or Data Scientist : uses the data that are already in the DB to analyze it and get insights and reports from it.

Note :

- Null => unknown , it's NOT Zero
- DBMS may be incompatible with any other DBMS (Microsoft solved this problem using XML files)

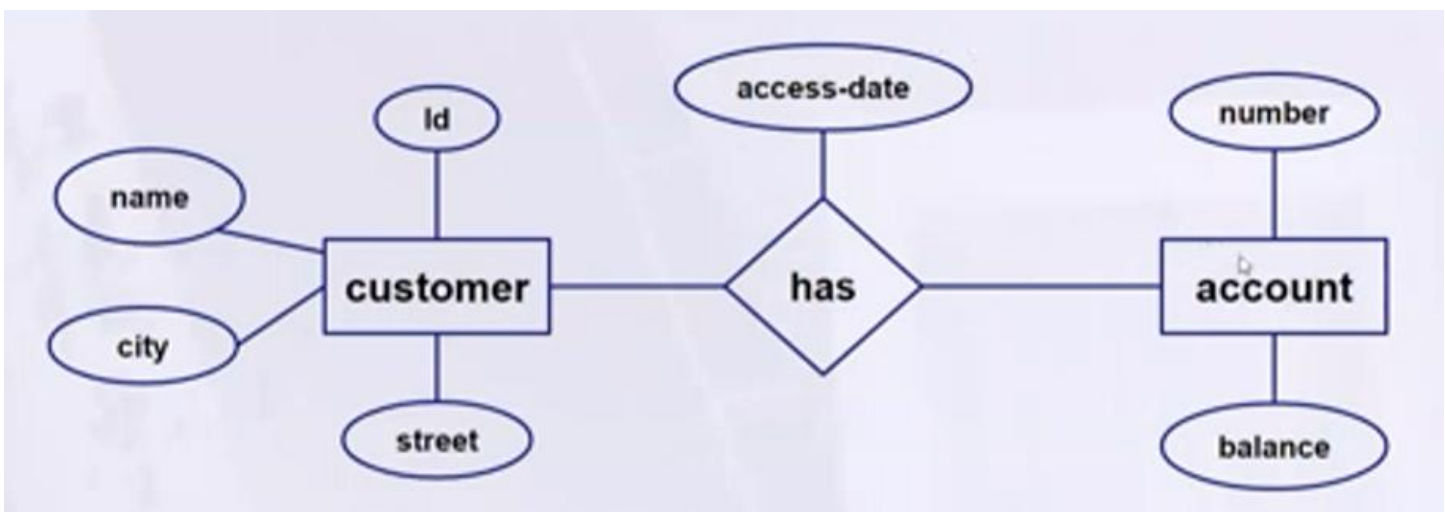


ERD (Entity relationship diagram) :

- Identifies information required by the business by displaying the relevant entities and the relationships between them.
- Basic Constructs of the ER model :
 - 1- Entities : Often Corresponds to a real time object that is distinguishable from any other object. ex: person , place , object , event (ex: holiday) ... [drawn as Rectangle]
 - 2- Attributes : Often corresponds to a field in the table. They are properties or characteristics of any entity. ex: name, location, [drawn as Ellipse]
 - 3- Relationships : corresponds to PK-FK equivalences in related tables. Link between entities [drawn as Diamond between the entities]

Note :

- Each entity must have one or more attributes.
- Foreign Key (FK) is not shown in the ERD , it's shown only in the Mapping Schema (actual tables)
- Usually , Nouns => Entities OR Attributes , Verbs => Relationships
- Each requirement document can have multiple ERDs , but each ERD has only one Schema (mapping)



Entities: Strong VS Weak

Strong entity :

- 1- It doesn't depend on other entity.
- 2- It has a primary key.

Weak entity : represented with Entity => DOUBLE RECTANGLE , Relationship => DOUBLE DIAMOND

- 1- It depends on other strong entity. ex: if we delete the parent (strong entity) , the child also will be deleted (weak entity)

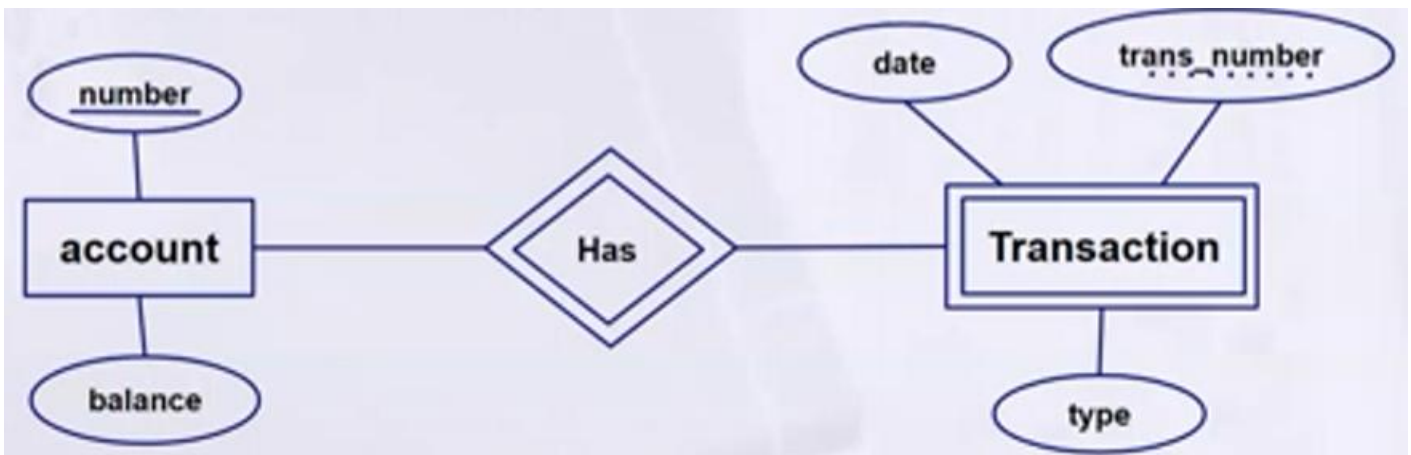
- 2- Doesn't have a primary key because it doesn't have sufficient attributes to form a primary key , so it has a partial key (depends on an employee).
- 3- Always Total Participation with the strong entity (will be discussed later in this doc.)

Primary Key :

- Uniquely identify the record
- Represented in an ellipse (as any attribute) and underlined by a SOLID line.

Partial Key :

- A set of attributes that can be associated with the PK of an owner entity set to distinguish a weak entity
- Represented in an ellipse (as any attribute) and underlined by a DOTTED line.

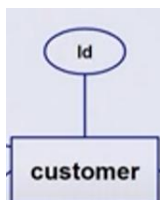


Ex: weak entity and strong entity

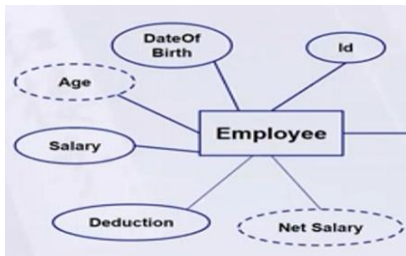
Attributes :

- Simple
- Derived
- Composite
- Multi-valued
- Complex

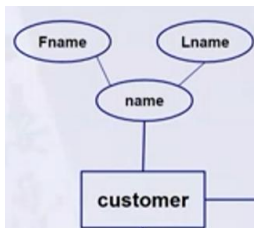
- 1- Simple attribute : default ellipse , ex: Id



- 2- Derived (Computed) attribute : dashed ellipse, can be generated while runtime , ex: Age



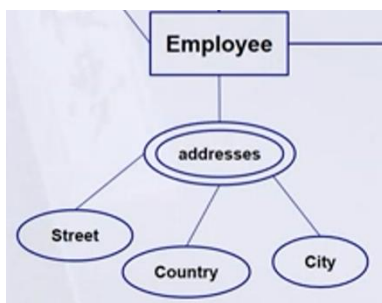
3- Composite attribute : ex: name



4- Multi-valued attribute : can have multiple values (Double ellipse) , ex: addresses

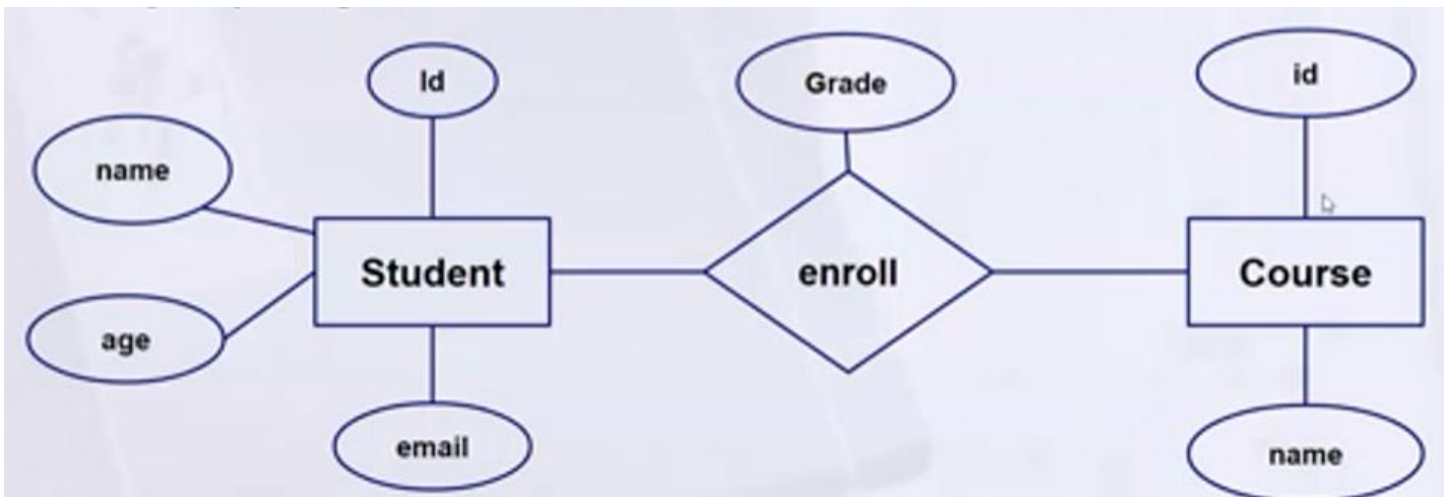


5- Complex attribute : multi-valued + composite , ex: addresses



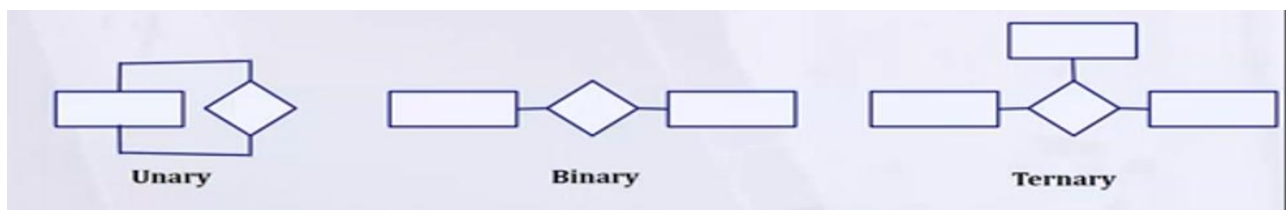
Relationships :

- Association between several entities.
- Relationship is between entities only , not entities and attributes or attributes and attributes
- We can have more than one relationship between same entities .
- We can have attributes on relationships



Three properties of relationships :

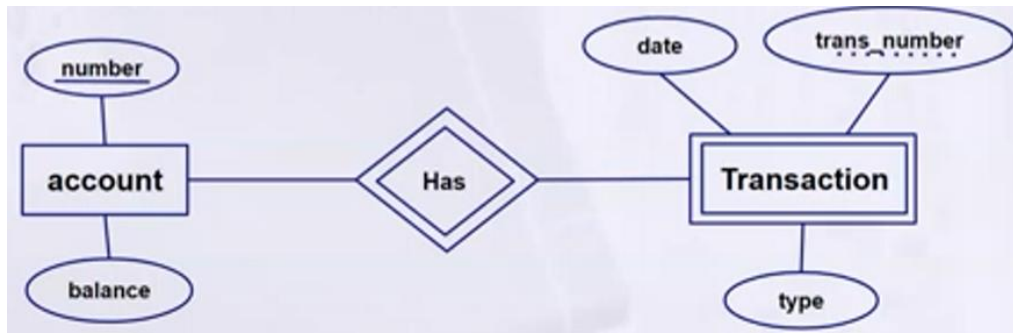
- 1- Degree of relationship : number of entities that participate in a relationship
 - Unary (Recursive/Self) : between two instances of one entity type (same entity).
 - Binary : between the instances of two entity types
 - Ternary : among the instances of three entity types



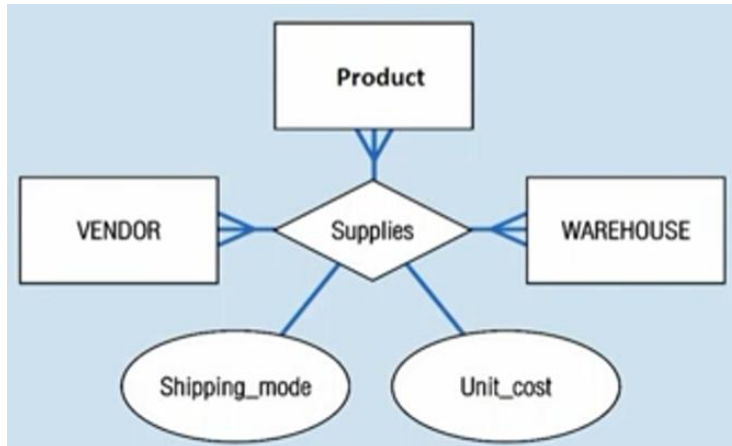
Unary : A relationship in which the same entity participates more than once



Binary : A set of degree 2

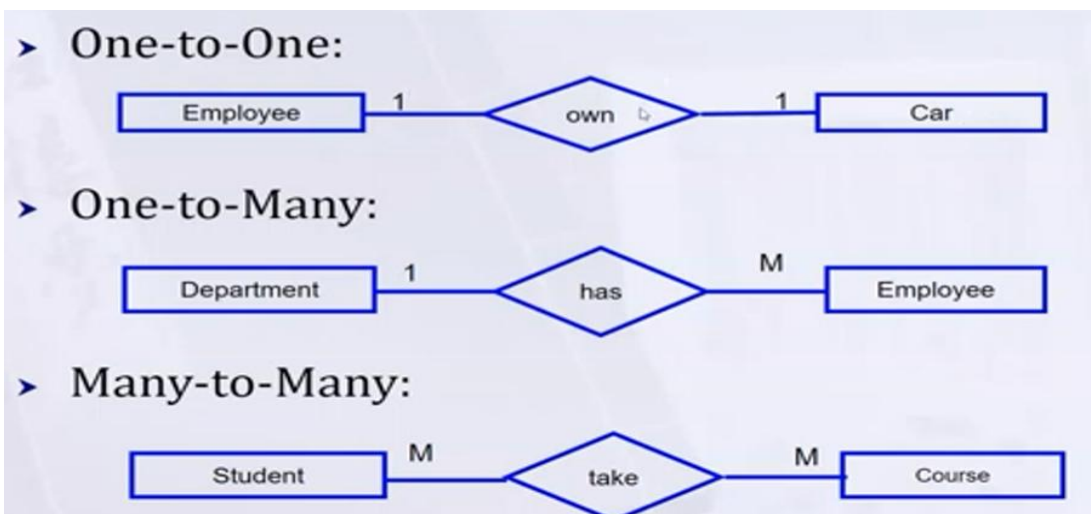


Ternary : A set of degree 3



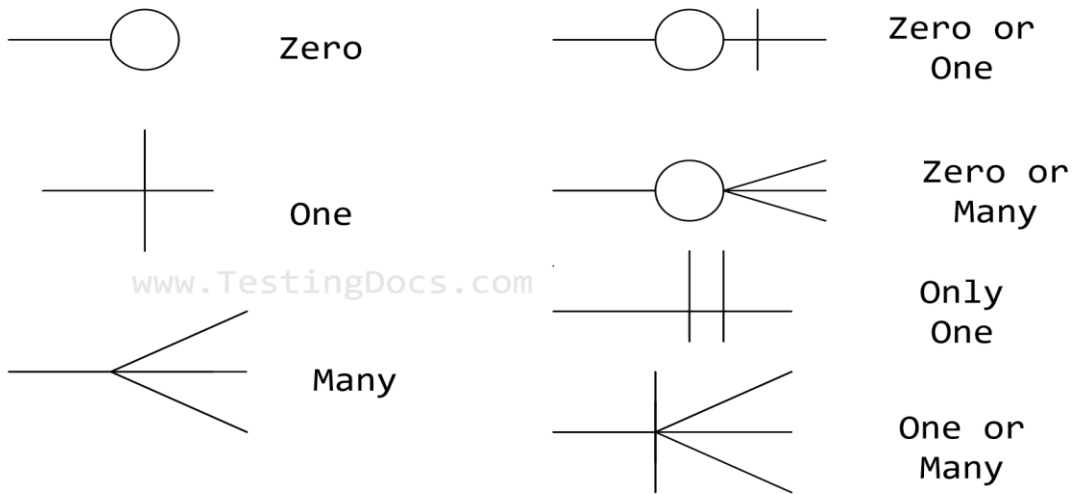
2- Cardinality constraint : How many instances of one entity will or must be connected to a single instance from the other entity.

- One – One relationship
- One – Many relationship
- Many – Many relationship



Another form

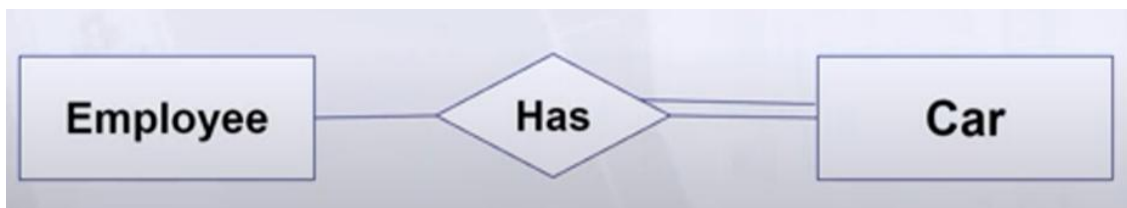
Cardinality Symbols



3- Participation constraint :

- Total (Mandatory) (double lines) : An employee **MUST** work for a department.
- Partial (Optional) (one line) : only **SOME** employees manage departments.

Ex1: an employee may have a car, a car may be assigned to an employee



Ex2: A department may hire many employees, an employee must be employed by a department



Note :

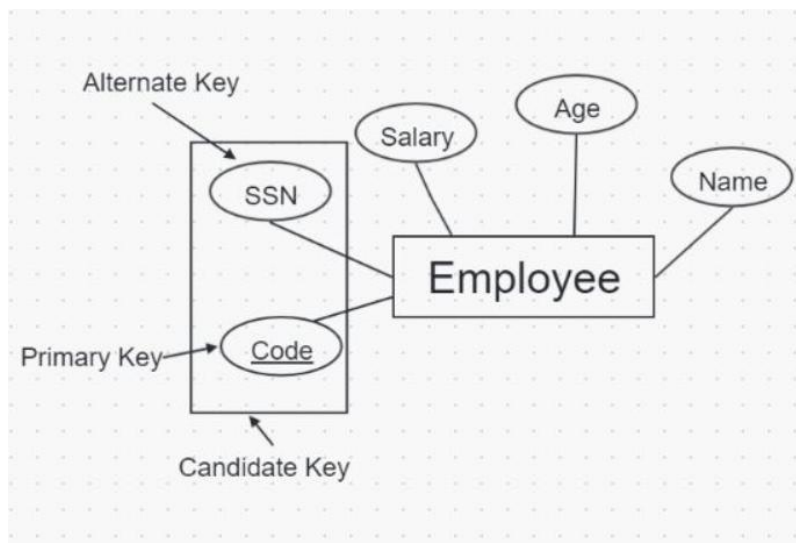
- Every weak entity is total participation with the strong entity , BUT NOT EVERY total participation a weak entity.
- It's NOT important to know the participation of Many-Many Relationships and Ternary Relationships , because they will be mapped in only one way (Will be discussed later)

Keys :

- 1- Primary key (identifier of the table) : unique + Don't allow NULLs
- 2- Candidate key : Keys that can be a primary key.
- 3- Foreign key : A key that comes from a foreign table (values from other table) + p.k in another table (this is not always right, violated in the self relationship , the PK and FK are in the same table)
- 4- Composite key : primary key of more than one column.
- 5- Partial key : key for a weak entity (underlined by a dotted line) .
- 6- Alternate key : keys that will remain from candidate keys after taking the primary key.
- 7- Super key : The set of keys that identify a row uniquely , candidate keys are subset of super keys
- 8- Compound key : Compound keys are always made up of two or more primary keys from other tables, a compound key is created when two or more primary keys from different tables are present as foreign keys within an entity. The foreign keys are used together to uniquely identify each record.

Note :

- Using a numerical PK is better than using a PK of type string
- We try to find a combination of attributes that can be a PK (unique) , if we cannot find any combination then we will create a new column ID



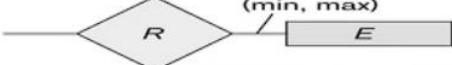
Self Study :

Associative entity

Structural constraint on participation min, max in ERD

Summary of notation for ER diagrams

Figure 3.14
Summary of the
notation for ER
diagrams.

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Identifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
	Total Participation of E_2 in R
	Cardinality Ratio 1 : N for $E_1:E_2$ in R
	Structural Constraint (min, max) on Participation of E in R